

ASSESSMENT TOOL 3 – Transformer Mechanics Worksheet

Course Code: CSA6502 **Course Title:** Generative AI and Large Scale Model

CO1 Statement: Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models.

Student Name: K. REVANTH ROYAL

Reg. No: 192472116

Section / Batch: SLOT-A

Date: 21-07-2026

Code of Conduct

I K. REVANTH ROYAL(192472116) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *K. REVANTH ROYAL*

1.Tokenization

(a) Split "DISAGREEMENT" into 3 sub-word pieces.

Answer:

- DIS
- AGREE
- MENT

Justification:

I split the word into DIS, AGREE, and MENT because each part has its own meaning. DIS is a prefix that means "not", AGREE is the main word, and MENT is a suffix that changes the word into a noun. This makes it easier for the model to understand the meaning of the whole word.

(b) Why does sub-word tokenization help?

Sub-word tokenization helps the model understand new words by breaking them into smaller parts that it already knows. Even if the model has never seen the word DISAGREEMENT, it can still understand it by combining the meanings of DIS, AGREE, and MENT.

2.Embeddings

(a) Which pair has higher cosine similarity? Why?

Answer: Dog and Puppy

Reason:

The words dog and puppy are closely related because a puppy is a young

dog. Their meanings are similar, so their embedding vectors are also close together. Dog and truck are completely different, so their similarity is much lower.

(b) What is an embedding? How is it different from a one-hot vector?

An embedding is a group of numbers that represents the meaning of a word. Words with similar meanings usually have similar embeddings.

A one-hot vector only shows the position of a word in the vocabulary. It cannot show any relationship between words like embeddings can.

3.Self-Attention

Sentence:

"The city council refused the protesters a permit because they feared violence."

(a) "They" refers to:

City council

(b) Why is self-attention needed?

Self-attention looks at the whole sentence to understand the meaning. In this sentence, the city council is the group that is most likely to fear violence, so the word "they" refers to the city council. This helps the model understand the sentence correctly instead of just choosing the nearest word.

4.Query, Key, and Value

Answer:

- Query (Q): It represents what the current word is looking for.
- Key (K): It represents the information each word provides.